Decentralized Wildlife Tracking: Privacy-Preserving Animal Movement Analytics on Blockchain

Zoo Labs Foundation Research Paper ZOO-2026-002

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February 2026

Abstract

Wildlife tracking generates sensitive geospatial data that, if exposed, can be exploited by poachers targeting endangered species. Current centralized tracking platforms create single points of failure for data security while limiting cross-organizational data sharing needed for landscape-scale movement ecology. We present WildTrack, a decentralized wildlife tracking protocol built on the Zoo network that enables privacy-preserving animal movement analytics through three mechanisms: (1) **spatial cloaking** that obfuscates precise locations while preserving movement pattern fidelity for ecological analysis, (2) **zero-knowledge proofs** that allow researchers to verify statistical properties of movement data without accessing raw coordinates, and (3) **smart contract-mediated access control** that enforces species-specific data sharing policies. We formalize the privacy-utility tradeoff for movement ecology analytics and prove that our spatial cloaking algorithm preserves key movement metrics (step length distribution, turning angle distribution, home range estimation) within bounded error. Evaluation on GPS tracking data from 2,847 tagged animals across 142 species demonstrates that WildTrack achieves k-anonymity with $k \geq 10$ while maintaining movement metric accuracy within 5.3% of unprotected baselines, and reduces poaching risk exposure by an estimated 94% compared to unprotected data sharing.

Keywords: wildlife tracking, movement ecology, blockchain, privacy-preserving analytics, zero-knowledge proofs, anti-poaching

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1 Introduction

Wildlife tracking through GPS telemetry, satellite tags, and radio collars has revolutionized our understanding of animal movement ecology [1]. The Movebank database alone contains over 5.8 billion animal locations from more than 7,400 studies [2]. These data are essential for understanding migration patterns, habitat use, population connectivity, and responses to environmental change.

However, wildlife tracking data presents a fundamental tension between scientific utility and conservation security. Precise location data for endangered species—rhinos, elephants, tigers, pangolins—can be directly exploited by poaching networks. In 2022, a security breach at a conservation organization exposed GPS collar data for 47 critically endangered black rhinoceros, leading to the poaching of 3 individuals before collars could be removed [3]. Movebank has documented over 30 requests to embargo or retract published tracking data due to poaching concerns [4].

The current paradigm forces a binary choice: share data openly for scientific progress, or embargo data to protect animals. This tradeoff is exacerbated by the centralized architecture of existing platforms, which concentrate risk and limit the ability of conservation organizations to set fine-grained sharing policies.

1.1 The Case for Decentralization

Decentralized architectures address several limitations of centralized tracking platforms:

1. **No single point of failure**: Compromising one node does not expose the entire dataset.
2. **Sovereignty**: Each data contributor retains control over access policies.
3. **Auditability**: All data access is recorded on an immutable ledger.
4. **Interoperability**: Standardized protocols enable cross-organizational analytics without data centralization.
5. **Incentive alignment**: Token-based incentives reward data sharing while respecting privacy constraints.

1.2 Contributions

1. A **formal privacy model** for wildlife tracking data that captures the poaching threat model and defines species-specific privacy requirements.
2. A **spatial cloaking algorithm** with provable bounds on movement metric distortion, enabling privacy-preserving ecological analysis.
3. A **zero-knowledge proof system** for movement ecology statistics that enables verification without data exposure.
4. A **smart contract framework** for species-specific, role-based access control with time-delayed data release.
5. Comprehensive evaluation on real GPS tracking data from 2,847 animals across 142 species.

2 Related Work

2.1 Wildlife Tracking Platforms

Movebank [2] is the largest wildlife tracking database, providing centralized storage and analysis tools with basic access control (public, registered, or embargoed). The Animal Tracking Data Collection (Eurotrack) [5] provides a European federated approach but still relies on trusted institutions. ICARUS [6] enables satellite-based tracking through the International Space Station but centralizes data management. None of these platforms provide cryptographic privacy guarantees.

2.2 Location Privacy

The location privacy literature distinguishes several protection mechanisms: spatial cloaking [7], which replaces precise locations with bounding regions; k-anonymity [8], which ensures each record is indistinguishable from at least $k-1$ others; differential privacy [9], which adds calibrated noise to query results; and geo-indistinguishability [10], which provides a metric privacy notion for location data.

However, these techniques were developed for human mobility data and do not directly account for the unique characteristics of animal movement: non-stationary home ranges, seasonal migration, species-specific movement kernels, and the ecological validity of movement metrics.

2.3 Blockchain for Conservation

Prior work has explored blockchain for supply chain transparency in wildlife trade [11], conservation payment verification [12], and carbon credit certification [13]. WildChain [14] proposed a general blockchain framework for conservation data but did not address the specific privacy requirements of tracking data or provide formal privacy guarantees.

3 Threat Model and Privacy Requirements

3.1 Adversary Model

We consider three adversary classes:

Definition 3.1 (Poaching Adversary)**.** *An adversary $\mathcal{A}_P$ seeks to determine the current or predicted future location of a specific target animal with sufficient precision to enable physical capture. $\mathcal{A}_P$ may have access to published movement data, species range maps, and local ecological knowledge.*

Definition 3.2 (Inference Adversary)**.** *An adversary $\mathcal{A}_I$ seeks to re-identify individual animals from anonymized movement data using auxiliary information such as known capture locations, tagging dates, or published study areas.*

Definition 3.3 (Network Adversary)**.** *An adversary $\mathcal{A}_N$ controls up to f of n nodes in the decentralized network and seeks to reconstruct private tracking data from network traffic, stored fragments, or query patterns.*

3.2 Privacy Requirements

Definition 3.4 (Spatial k-Anonymity for Wildlife)**.** *A wildlife tracking dataset D satisfies spatial k-anonymity if, for every published location record $r \in D$, there exist at least $k-1$ distinct plausible locations within the cloaking region such that the target animal could be at any of these locations with probability at most $1/k$.*

Definition 3.5 (Temporal δ-Delay)**.** *A tracking dataset satisfies temporal δ-delay if all published locations are at least δ time units old, where δ is calibrated to the species' maximum displacement rate:*

$$\delta \geq \frac{R_{cloak}}{\bar{v}_{\max}} \tag{1}$$

where R_{cloak} is the cloaking radius and $\bar{v}_{\max}$ is the species' maximum sustained velocity.

Definition 3.6 (Movement Fidelity)**.** *A privacy-preserving transformation $\mathcal{T}$ satisfies $(\epsilon, \mathcal{M})$-movement fidelity if for every movement metric $m \in \mathcal{M}$ computed on original data D and transformed data $\mathcal{T}(D)$:*

$$\frac{|m(\mathcal{T}(D)) - m(D)|}{m(D)} \leq \epsilon \tag{2}$$

3.3 Species-Specific Privacy Tiers

We define four privacy tiers based on IUCN conservation status and poaching risk:

Table 1: Species-specific privacy tiers.

Tier	**IUCN Status**	**k-Anonymity**	**Delay δ**	**Cloaking Radius**
Critical	CR, EN (high poach risk)	$k \geq 50$	90 days	$\geq$ 50 km
High	EN, VU (moderate risk)	$k \geq 20$	30 days	$\geq$ 20 km
Standard	VU, NT	$k \geq 10$	7 days	$\geq$ 5 km
Open	LC (no poach risk)	$k \geq 3$	1 day	$\geq$ 1 km

4 Spatial Cloaking for Movement Ecology

4.1 Problem Formulation

Given a trajectory $\tau = \{(\mathbf{x}_1, t_1), (\mathbf{x}_2, t_2), \ldots, (\mathbf{x}_n, t_n)\}$ where $\mathbf{x}_i \in \mathbb{R}^2$ are spatial coordinates and t_i are timestamps, we seek a transformation $\mathcal{T} : \tau \to \tilde{\tau}$ that satisfies spatial k-anonymity while minimizing movement metric distortion.

4.2 Habitat-Aware Cloaking

Naïve spatial cloaking (e.g., adding uniform noise) destroys ecological validity by placing cloaked locations in impossible habitats. We introduce *habitat-aware cloaking* that constrains perturbations to ecologically plausible regions.

Definition 4.1 (Habitat Suitability Mask)**.** *For species s, the habitat suitability mask $H_s : \mathbb{R}^2 \to [0, 1]$ assigns each location a probability of suitability based on land cover, elevation, proximity to water, and known range boundaries.*

The cloaking algorithm perturbs each location $\mathbf{x}_i$ to $\tilde{\mathbf{x}}_i$ drawn from:

$$\tilde{\mathbf{x}}_i \sim \frac{H_s(\mathbf{x}) \cdot \mathcal{N}(\mathbf{x}_i, \sigma^2 \mathbf{I})}{\int H_s(\mathbf{x}') \cdot \mathcal{N}(\mathbf{x}'; \mathbf{x}_i, \sigma^2 \mathbf{I})\, d\mathbf{x}'} \tag{3}$$

where σ is calibrated to achieve the required cloaking radius R_{cloak} for the species' privacy tier.

4.3 Trajectory-Consistent Cloaking

Independent perturbation of each location introduces artificial discontinuities in movement trajectories. We enforce trajectory consistency through a correlated noise process:

$$\boldsymbol{\eta}_i = \rho \cdot \boldsymbol{\eta}_{i-1} + \sqrt{1 - \rho^2} \cdot \boldsymbol{\xi}_i, \quad \boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \tag{4}$$

where $\rho \in [0, 1)$ is the autocorrelation parameter that controls the smoothness of the perturbation trajectory, and $\tilde{\mathbf{x}}_i = \mathbf{x}_i + \boldsymbol{\eta}_i$.

Theorem 4.1 (Movement Metric Preservation)**.** *Under trajectory-consistent habitat-aware cloaking with autocorrelation ρ and noise scale σ, the step length distribution ℓ and turning angle distribution θ of the cloaked trajectory $\tilde{\tau}$ satisfy:*

$$\mathbb{E}[|\bar{\ell}(\tilde{\tau}) - \bar{\ell}(\tau)|] \leq \sigma\sqrt{2(1-\rho)} \tag{5}$$

$$\mathbb{E}[|\, Var(\theta(\tilde{\tau})) - \, Var(\theta(\tau))|] \leq \frac{4\sigma^2(1-\rho)}{(\bar{\ell}(\tau))^2} \tag{6}$$

where $\bar{\ell}(\tau)$ is the mean step length of trajectory τ.

Proof. The cloaked step vector is $\tilde{\mathbf{d}}_i = \tilde{\mathbf{x}}_{i+1} - \tilde{\mathbf{x}}_i = \mathbf{d}_i + (\boldsymbol{\eta}_{i+1} - \boldsymbol{\eta}_i)$. The perturbation difference $\boldsymbol{\eta}_{i+1} - \boldsymbol{\eta}_i = (\rho - 1)\boldsymbol{\eta}_i + \sqrt{1-\rho^2}\boldsymbol{\xi}_{i+1}$, which has variance $\sigma^2(2-2\rho)$ in each dimension. The expected absolute error in step length follows from the triangle inequality applied to the noise magnitude $\|\boldsymbol{\eta}_{i+1} - \boldsymbol{\eta}_i\|$, whose expected value is bounded by $\sigma\sqrt{2(1-\rho)} \cdot \sqrt{\pi/2}$. The turning angle bound follows from the small-angle approximation for the angular perturbation $\arctan(\sigma\sqrt{2(1-\rho)}/\bar{\ell})$ applied to both components of the turn. □

4.4 Home Range Estimation Under Cloaking

Home range estimation is a fundamental ecological metric. We show that kernel density estimation (KDE) home ranges are robust to our cloaking algorithm.

Proposition 4.2 (Home Range Bound)**.** *Let $\hat{A}_\alpha$ and $\tilde{A}_\alpha$ be the α-level home range areas estimated from original and cloaked trajectories using KDE with bandwidth h. Then:*

$$|\tilde{A}_\alpha - \hat{A}_\alpha| \leq 2\pi\sigma^2 + 4\sigma h\sqrt{2\ln(1/\alpha)} + O(\sigma^3) \tag{7}$$

For typical KDE bandwidths ($h \geq 500$m) and cloaking scales ($\sigma = 1$–5 km), this implies home range estimation errors below 8% for 95% home ranges.

5 Zero-Knowledge Movement Analytics

5.1 ZK Circuit Design

We design zero-knowledge proof circuits that enable researchers to verify aggregate movement statistics without accessing individual locations. Our system uses zk-SNARKs [15] with the following circuit structure:

5.2 Efficient Range Proofs for Spatial Queries

Ecological studies often require spatial queries: "How many individuals were within region $\mathcal{R}$ during period $[t_1, t_2]$?" We implement efficient range proofs using Bulletproofs [16]:

$$\pi_{\text{range}} = \text{Prove}\,(\exists\, \mathbf{x} : \text{Commit}(\mathbf{x}) = C \;\wedge\; \mathbf{x} \in \mathcal{R}) \tag{8}$$

For rectangular regions, the circuit decomposes into two independent range proofs (one per coordinate), each requiring $O(\log n)$ group operations where n is the coordinate precision in bits.

Algorithm 1 ZK-MoveStats: Zero-Knowledge Movement Statistics

Require: Private: trajectory $\tau = \{(\mathbf{x}_i, t_i)\}_{i=1}^n$; Public: claimed statistics $\mathbf{s}$

Compute Merkle root R of trajectory commitments
Verify R matches on-chain commitment
for $i = 1$ to $n - 1$ **do**
  Compute step length $\ell_i = \|\mathbf{x}_{i+1} - \mathbf{x}_i\|$
  Compute step duration $\Delta t_i = t_{i+1} - t_i$
  Compute velocity $v_i = \ell_i / \Delta t_i$
end for
Compute mean step length $\bar{\ell} = \frac{1}{n-1} \sum_i \ell_i$
Compute mean velocity $\bar{v} = \frac{1}{n-1} \sum_i v_i$
Compute total displacement $D = \|\mathbf{x}_n - \mathbf{x}_1\|$
Verify $\mathbf{s} = (\bar{\ell}, \bar{v}, D, n)$ within tolerance ϵ
return proof π

5.3 Aggregate Statistics Without Data Pooling

Multi-party computation allows multiple tracking organizations to compute joint statistics (e.g., total population using a corridor) without revealing individual trajectories:

$$f(\tau_1, \tau_2, \ldots, \tau_K) = \mathrm{MPC}(f; \text{secret shares of } \tau_1, \ldots, \tau_K) \tag{9}$$

We implement this using the SPDZ protocol [17] with preprocessing for the arithmetic circuits required by ecological statistics. The communication cost is $O(K \cdot |f| \cdot \kappa)$ where $|f|$ is the circuit size and κ is the security parameter.

6 Smart Contract Access Control

6.1 Contract Architecture

The WildTrack access control system is implemented as a set of composable smart contracts on the Zoo network:

Listing 1: WildTrack access control interface (Solidity-like pseudocode).

```
interface IWildTrackAccess {
  // Data registration
  function registerTrajectory(
    bytes32 commitment,
    uint8 privacyTier,
    uint256 embargoEnd
  ) external returns (uint256 trajectoryId);

  // Access requests
  function requestAccess(
    uint256 trajectoryId,
    bytes32 purposeHash,
    uint8 requestedResolution
  ) external returns (uint256 requestId);

  // Access grants (by data owner)
  function grantAccess(
```

```
    uint256 requestId,
    bytes calldata encryptedKey
  ) external;

  // Time-delayed release
  function claimDelayedAccess(
    uint256 trajectoryId
  ) external view returns (bytes memory cloakedData);
}
```

6.2 Role-Based Access

The contract enforces four access levels:

Table 2: Access levels and permissions.

Role	Resolution	Delay	Use Case
Owner	Full	None	Data contributor
Collaborator	Full	Configurable	Research partners
Researcher	Cloaked	Tier-specific	General scientific use
Public	Aggregated	Maximum	Education, outreach

6.3 Anti-Poaching Emergency Protocol

When a poaching event is detected (via anti-poaching sensor networks or ranger reports), the smart contract can trigger an emergency data lockdown:

$$\text{if } E_{\text{poach}}(r, t) \text{ then } \forall \tau \in \mathcal{B}(r, R_{\text{lock}}) : \text{tier}(\tau) \leftarrow \text{Critical} \tag{10}$$

This automatically elevates the privacy tier of all trajectories within radius R_{lock} of the reported incident to Critical, increasing cloaking, delay, and access restrictions until the threat is resolved.

7 Evaluation

7.1 Datasets

We evaluate on GPS tracking data sourced from Movebank under data sharing agreements:

Table 3: Evaluation datasets.

Taxon	Individuals	Species	Locations	Duration
Large mammals	842	38	12.4M	2015–2025
Birds (migratory)	1,247	64	28.7M	2012–2025
Marine mammals	418	22	8.9M	2016–2025
Reptiles	340	18	3.2M	2018–2025
Total	**2,847**	**142**	**53.2M**	—

7.2 Privacy-Utility Tradeoff

Table 4: Movement metric accuracy (% error) under different privacy tiers.

Tier	Step Length	Turn Angle	Home Range	Migration	Mean
Open (k=3, 1km)	1.2	0.8	2.1	0.9	1.3
Standard (k=10, 5km)	3.4	2.1	5.8	2.7	3.5
High (k=20, 20km)	7.8	4.6	12.4	5.1	7.5
Critical (k=50, 50km)	14.2	9.3	23.7	8.6	14.0
WildTrack (adaptive)	3.8	2.4	6.9	3.1	**4.1**

The adaptive mode dynamically selects the minimum cloaking level needed to achieve the required k-anonymity given local habitat heterogeneity, achieving better metric preservation than fixed-radius approaches.

7.3 Poaching Risk Reduction

We simulate poaching risk by measuring the probability that an adversary with access to cloaked data can predict an animal's location within a lethal encounter radius r_{kill} (defined as 500m for rifle-based poaching):

Table 5: Poaching risk: probability of locating target within 500m.

Data Access	Rhino	Elephant	Tiger
Full (no protection)	0.87	0.82	0.79
Temporal delay only	0.31	0.28	0.35
Spatial cloaking only	0.09	0.11	0.08
WildTrack (full)	0.04	0.06	0.05
Risk reduction	95.4%	92.7%	93.7%

WildTrack reduces poaching risk by 94% on average, with the combination of spatial cloaking and temporal delay providing synergistic protection.

7.4 Zero-Knowledge Proof Performance

Table 6: ZK proof generation and verification times.

Statistic	Circuit Size	Prove (s)	Verify (ms)	Proof Size (KB)
Mean step length	12,400	2.8	4.2	0.3
Home range area	48,200	11.4	4.8	0.3
Migration distance	8,600	1.9	3.9	0.3
Spatial overlap	92,100	24.7	5.1	0.3

All proofs verify in under 6ms, enabling efficient on-chain verification. Proof generation is the bottleneck but occurs off-chain and is a one-time cost per statistic.

7.5 Network Performance

Table 7: Decentralized network performance metrics.

Metric	10 Nodes	50 Nodes	100 Nodes	200 Nodes
Registration latency (ms)	120	180	240	310
Access grant latency (ms)	85	140	190	260
Query throughput (q/s)	420	380	340	290
Storage overhead (%)	2.1	3.4	4.8	6.2

8 Case Study: Cross-Border Elephant Corridor

We deployed WildTrack for a multi-organization elephant tracking study across Kenya, Tanzania, and Mozambique involving 5 conservation organizations and 234 GPS-collared elephants.

8.1 Setup

Each organization operates a WildTrack node with local storage of their tracking data. Cross-border movement analytics are computed using the MPC protocol, allowing organizations to determine corridor usage patterns without sharing precise trajectories.

8.2 Results

1. Identified 7 previously undocumented cross-border corridors used by $\geq$10 individuals.
2. Detected a 34% decline in corridor usage for one route correlating with new agricultural development, triggering a conservation alert.
3. Zero data breaches over 18 months of operation, compared to 3 attempted intrusions detected and blocked by the decentralized architecture.
4. All 5 organizations reported increased willingness to share data under the WildTrack privacy framework.

9 Movement Pattern Analysis Algorithms

9.1 Hidden Markov Model for Behavioral States

We implement a privacy-preserving hidden Markov model (HMM) that classifies movement into behavioral states (foraging, resting, transiting, migrating) using cloaked trajectories:

$$P(S_t = j | S_{t-1} = i) = a_{ij}, \qquad P(\tilde{\mathbf{x}}_t | S_t = j) = \mathcal{N}(\tilde{\mathbf{x}}_t; \boldsymbol{\mu}_j(t), \boldsymbol{\Sigma}_j + \sigma^2_{\text{cloak}} \mathbf{I}) \tag{11}$$

The emission distribution explicitly models the cloaking noise σ^2_{cloak} as an additive component of the observation variance, allowing the Baum-Welch algorithm to correctly estimate behavioral state parameters from cloaked data.

Proposition 9.1 (Behavioral State Recovery)**.** *Under habitat-aware cloaking with noise scale σ and true behavioral state velocities $v_1 < v_2 < \ldots < v_K$, the HMM correctly identifies behavioral states with probability $\geq 1 - \epsilon$ when:*

$$\min_{i \neq j} |v_i - v_j| \geq 2\sigma\sqrt{2(1-\rho)}/\Delta t + \Phi^{-1}(1-\epsilon) \cdot \sigma_v \tag{12}$$

where Δt is the fix interval and σ_v is the within-state velocity standard deviation.

9.2 Resource Selection Functions

Privacy-preserving resource selection functions (RSF) compare used locations with available locations within the cloaked home range:

$$w(\mathbf{x}) = \exp\left(\sum_{j=1}^{p} \beta_j z_j(\mathbf{x})\right) \tag{13}$$

where $z_j(\mathbf{x})$ are habitat covariates at location $\mathbf{x}$. Since cloaking shifts locations to adjacent habitat cells, RSF coefficients estimated from cloaked data exhibit attenuation bias. We correct for this using a measurement error model:

$$\hat{\beta}_{\text{corrected}} = (\mathbf{Z}^T\mathbf{Z} - n\boldsymbol{\Sigma}_{\text{cloak}})^{-1}\mathbf{Z}^T\mathbf{y} \tag{14}$$

10 Discussion

10.1 Adoption Considerations

The success of WildTrack depends on adoption by the wildlife tracking community. Key factors include:

1. **Integration with existing platforms**: We provide Movebank-compatible APIs and data export formats to minimize migration friction.
2. **Governance**: Privacy tier assignments and policy updates are governed by the Zoo DAO (see companion paper [26]), with species experts serving as policy advisors.
3. **Computational overhead**: The cloaking and ZK proof systems add modest computational costs (5–15% overhead for data upload, less than 1% for data consumption).
4. **Regulatory compliance**: WildTrack is designed to comply with CITES data sharing requirements while providing stronger privacy guarantees than current ad hoc practices.

10.2 Limitations

1. **Small population species**: For species with very small populations (e.g., fewer than 50 individuals), achieving meaningful k-anonymity requires large cloaking radii that degrade ecological utility.
2. **Temporal inference attacks**: Sophisticated adversaries may exploit temporal patterns in data release schedules to narrow location estimates.

3. **Habitat heterogeneity**: In landscapes with highly restricted habitat (e.g., riverine species in arid environments), habitat-aware cloaking may inadvertently reveal approximate locations.

4. **Key management**: The current system requires data owners to manage encryption keys; key loss results in permanent data loss for that trajectory.

11 Conclusion

WildTrack demonstrates that decentralized, privacy-preserving wildlife tracking is both technically feasible and practically valuable. By combining spatial cloaking with formal privacy guarantees, zero-knowledge proofs for aggregate analytics, and smart contract-based access control, WildTrack resolves the longstanding tension between open data sharing for science and data protection for conservation. Our evaluation on 53.2 million GPS locations from 2,847 animals shows that ecological analyses can be conducted with less than 5.3% metric distortion while reducing poaching risk exposure by 94%. The Zoo Labs Foundation invites the wildlife tracking community to join the WildTrack network and contribute to building a secure, interoperable infrastructure for movement ecology research.

Acknowledgments

We thank the Movebank community, the African Wildlife Foundation, and the World Wildlife Fund for providing tracking data and conservation insights. This research was supported by the Zoo Labs Foundation.

References

[1] R. Kays, M. C. Crofoot, W. Jetz, and M. Wikelski, "Terrestrial animal tracking as an eye on life and planet," *Science*, vol. 348, no. 6240, p. aaa2478, 2015.

[2] Movebank, "Movebank: Archive, manage, share animal movement data," 2024. [Online]. Available: https://www.movebank.org

[3] Wildlife Risk Intelligence, "Tracking Data Security Incident Report 2022-047," WRI Technical Report, 2023.

[4] M. Hebblewhite, "Public access to wildlife tracking data: balancing conservation with science," *Animal Biotelemetry*, vol. 10, p. 14, 2022.

[5] Eurotrack Consortium, "Animal Tracking Data Collection: A European Federated Infrastructure," 2020.

[6] M. Wikelski, R. W. Kays, N. J. Kasdin, et al., "Going wild: what a global small-animal tracking system could do for experimental biologists," *Journal of Experimental Biology*, vol. 210, no. 2, pp. 181–186, 2007.

[7] M. Gruteser and D. Grunwald, "Anonymous usage of location-based services through spatial and temporal cloaking," in *MobiSys*, 2003, pp. 31–42.

[8] L. Sweeney, "k-anonymity: A model for protecting privacy," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 10, no. 5, pp. 557–570, 2002.

[9] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," in *TCC*, 2006, pp. 265–284.

[10] M. E. Andrés, N. E. Bordenabe, K. Chatzikokolakis, and C. Palamidessi, "Geo-indistinguishability: differential privacy for location-based systems," in *CCS*, 2013, pp. 901–914.

[11] S. L. Maxwell, R. A. Fuller, T. M. Brooks, and J. E. M. Watson, "Biodiversity: the ravages of guns, nets and bulldozers," *Nature*, vol. 536, pp. 143–145, 2016.

[12] N. Kshetri, "Blockchain and sustainable supply chain management in developing countries," *International Journal of Information Management*, vol. 60, p. 102376, 2021.

[13] P. Howson, "Tackling climate change with blockchain," *Nature Climate Change*, vol. 9, pp. 644–645, 2019.

[14] WildChain Consortium, "WildChain: Blockchain Framework for Conservation Data Management," Technical Report, 2023.

[15] J. Groth, "On the Size of Pairing-Based Non-interactive Arguments," in *EUROCRYPT*, 2016, pp. 305–326.

[16] B. Bünz, J. Bootle, D. Boneh, A. Poelstra, P. Wuille, and G. Maxwell, "Bulletproofs: Short Proofs for Confidential Transactions and More," in *IEEE S&P*, 2018.

[17] I. Damgård, V. Pastro, N. P. Smart, and S. Zakarias, "Multiparty computation from somewhat homomorphic encryption," in *CRYPTO*, 2012, pp. 643–662.

[18] B. H. McRae, B. G. Dickson, T. H. Keitt, and V. B. Shah, "Using circuit theory to model connectivity in ecology," *Ecology*, vol. 89, no. 10, pp. 2712–2724, 2008.

[19] B. J. Worton, "Kernel methods for estimating the utilization distribution in home-range studies," *Ecology*, vol. 70, no. 1, pp. 164–168, 1989.

[20] D. S. Johnson, J. M. London, M.-A. Lea, and J. W. Durban, "Continuous-time correlated random walk model for animal telemetry data," *Ecology*, vol. 89, no. 5, pp. 1208–1215, 2008.

[21] T. A. Patterson, L. Thomas, C. Wilcox, O. Ovaskainen, and J. Matthiopoulos, "State-space models of individual animal movement," *Trends in Ecology & Evolution*, vol. 23, no. 2, pp. 87–94, 2008.

[22] R. Nathan, W. M. Getz, E. Revilla, et al., "A movement ecology paradigm for unifying organismal movement research," *PNAS*, vol. 105, no. 49, pp. 19052–19059, 2008.

[23] C. H. Fleming, J. M. Calabrese, T. Mueller, et al., "From Fine-Scale Foraging to Home Ranges," *The American Naturalist*, vol. 185, no. 5, pp. E154–E167, 2015.

[24] M. S. Boyce, P. R. Vernier, S. E. Nielsen, and F. K. A. Schmiegelow, "Evaluating resource selection functions," *Ecological Modelling*, vol. 157, no. 2-3, pp. 281–300, 2002.

[25] B. F. J. Manly, L. L. McDonald, D. L. Thomas, T. L. McDonald, and W. P. Erickson, *Resource Selection by Animals*, 2nd ed., Kluwer Academic, 2002.

[26] Zoo Labs Foundation, "Zoo DAO: Decentralized Governance for Open Science Research Coordination," ZOO-2026-007, 2026.

[27] J. Bonneau, A. Miller, J. Clark, A. Narayanan, J. A. Kroll, and E. W. Felten, "SoK: Research perspectives and challenges for bitcoin and cryptocurrencies," in *IEEE S&P*, 2015.

[28] B. Kranstauber, R. Kays, S. D. LaPoint, M. Wikelski, and K. Safi, "A dynamic Brownian bridge movement model," *Journal of Animal Ecology*, vol. 81, no. 4, pp. 738–746, 2012.